



Cite this: *Phys. Chem. Chem. Phys.*,
2017, **19**, 17366

Difference in chemical bonding between lithium and sodium salts: influence of covalency on their solubility†

Su Chen,^a Jun Ishii,^b Shunsuke Horiuchi,^b Masahiro Yoshizawa-Fujita^b and Ekaterina I Izgorodina  ^{✉a}

Recent thermal runaways in lithium-ion batteries have reinforced the focus on the research of safer electrolytes based on ionic liquids. A simple switch from organic solvents to ionic liquids has been proven difficult due to the decreased efficiency of batteries caused by decreased conductivity and increased viscosity of ionic liquids upon addition of lithium salts. The new trend in replacing lithium salts with a cheaper alternative, sodium salts, has resulted in rather poor solubility of sodium salts in commonly used ionic liquids. This phenomenon has been left largely unexplained. Herein, we present a high-level quantum chemical study of the chemical bonding of lithium and sodium salts coupled with ionic liquid anions. Due to their proximity to the anion, the $1s^2$ electrons on the lithium cation are found to become strongly polarized by the presence of the anion such that they start participating in the bonding, making it more covalent than originally thought. In sodium salts the $2s^2$ orbitals are rather removed from the anion, making its influence weaker. This polarization results in 90 kJ mol^{-1} of difference in the interaction magnitude between lithium and sodium salts. Theoretical results have confirmed that increasing covalency in lithium salts results in their excellent solubility since these dissolve as ion-paired complexes. The downside of this ability is decreased conductivity as lithium salts are unlikely to easily dissociate in ionic liquids. Sodium salts are shown to maintain a high degree of ionicity, thus decreasing their chances of being solvated by ionic liquids as a result of their low concentration of ions per unit volume. The theoretical results are further underpinned by solubility studies of MX salts, where $M = \text{Li or Na}$ and $X = \text{bis(trifluoromethylsulfonyl)imide (NTf}_2\text{), BF}_4^- \text{ or PF}_6^-$, conducted in six different ionic liquids. Lithium salts consisting of BF_4^- or PF_6^- exhibited significantly better solubility than their sodium analogues by at least an order of magnitude. The findings of this work will have implications on the future direction of the development of safe electrolytes for lithium and sodium-ion secondary batteries.

Received 7th May 2017,
Accepted 15th June 2017

DOI: 10.1039/c7cp03009f

rsc.li/pccp

Introduction

Recent years have seen a growing interest in enhancing the performance and longevity of lithium- and sodium-ion rechargeable batteries.^{1–9} The electrolyte inside the battery represents the main hazard because of its volatile organic nature that might eventually cause thermal runaway. The most

infamous incident of runaway occurred on the Boeing 787 Dreamliner that resulted in the grounding of its entire fleet.¹⁰

Ionic liquids consisting entirely of organic and inorganic ions have been heralded as replacement electrolytes in metal-ion batteries.^{8,9} Despite the fact that preliminary data have been rather promising, the main issue of completely switching to ionic liquids has been unresolved. The addition of lithium and sodium salts has faced an intriguing problem resulting in the decreased performance of ionic liquids – a significant decrease in conductivity, accompanied by an increase in viscosity.^{11,12} Both factors negatively affect the performance of the battery.^{7,13–15} The idea of replacing the costly lithium ions with a cheaper version in the form of sodium cations has experienced a largely unforeseen problem due to poor solubility of sodium salts in ionic liquids,^{5,16} apart from sodium bis(trifluoromethyl-sulfonyl)imide.¹⁷ It is even more puzzling

^a School of Chemistry, Monash University, 17 Rainforest Walk, Clayton, Victoria, 3800, Australia. E-mail: katya.pas@monash.edu

^b Department of Materials and Life Sciences, Sophia University, 7-1 Kioi-cho, Chiyoda-ku, Tokyo 102-8554, Japan

† Electronic supplementary information (ESI) available: Optimized structures of lithium and sodium salts, interionic distances, total and correlation electronic energies at CCSD(T) and MP2 levels of theory from full-core and frozen-core calculations, Geodesic net charge transfers and measured solubility data. See DOI: 10.1039/c7cp03009f

that the solubility of lithium salts in ionic liquids is excellent, despite a much higher charge density on the lithium cation, which would require a larger solvation shell than the sodium cation. The phenomenon of striking differences in the solubility of lithium and sodium salts in ionic liquids mentioned in a number of publications has not been explained. Moreover, from the theoretical point of view, the presence of non-physical positive correlation interaction energies was observed in M^+X^- complexes of fluorides and chlorides of lithium and sodium,¹⁸ which was largely left unexplained. It was hypothesized that the multi-reference character might be responsible for this unusual trend but never proven.

Experimental

Theoretical procedures

In this work a thorough study has been conducted into the bonding of LiX and NaX salts, with X representing commonly used ionic liquid anions such as bis(fluorosulfonyl)imide (FSI), tetrafluoroborates (BF_4^-), hexafluorophosphate (PF_6^-) and dicyanamide (DCA). The geometries of these salts were optimized in both the gas phase and using the conductor-like polarisable continuum model¹⁹ with chloroform ($CHCl_3$) and dimethylsulfoxide (DMSO) as solvents (see Fig. 1 in the ESI†). In the case of the FSI and PF_6^- anions, a few energetically favorable configurations were considered. Interaction energies were calculated using full core MP2 and CCSD(T) levels of theory combined with cc-pVTZ (denoted as VTZ) and aug-cc-pVTZ (denoted as AVTZ) basis sets and including counterpoise correction proposed by Boys and Bernardi.²⁰ The convergence of MP2 interaction energies to the complete basis set (CBS) was studied through the Helgaker extrapolation²¹ of (aug)-cc-pVTZ and (aug)-cc-VQZ for the structures optimized with DMSO as the solvent. Weakly polar chloroform was chosen as it resembles the dielectric constants of sodium inorganic salts, whereas highly polar DMSO mimics the ionic environment that a salt might experience in ionic liquids. Therefore, for the purpose of this study, DMSO optimized M^+X^- complexes represent the best model of an alkali metal salt in an ionic liquid. Net charge

transfer was predicted using atomic charges calculated with the Geodesic point selection scheme²² at the HF/aug-cc-pVTZ level of theory. All quantum chemical calculations were performed using the Gaussian 09,²³ PSI4²⁴ and GAMESS-US²⁵ packages.

Experimental procedures

Solubility studies of MX salts (where $M = Li$ and Na and $X = Ntf_2^-$, BF_4^- , and PF_6^-) were conducted in six ionic liquids such as 1-ethyl-3-methylimidazolium bis(fluorosulfonyl)imide ($[C_2mim][FSI]$), 1-ethyl-3-methylimidazolium bis(trifluoro-methylsulfonyl)imide ($[C_2mim][Ntf_2]$), N -methyl- N -propyl-pyrrolidinium bis(fluoro-sulfonyl)imide ($[C_3mpyr][FSI]$), N -methyl- N -propyl-pyrrolidinium bis(trifluoromethylsulfonyl)-imide ($[C_3mpyr][Ntf_2]$), N -methyl- N -methoxymethyl-pyrrolidinium bis(fluorosulfonyl)imide ($[C_{101}mpyr][FSI]$), and N -methyl- N -methoxymethyl-pyrrolidinium bis(fluorosulfonyl)-imide ($[C_{101}mpyr][Ntf_2]$). $[C_2mim][FSI]$, $[C_2mim][Ntf_2]$, and $[C_3mpyr][FSI]$ were purchased from Kanto Chemical Co., Inc. $[C_3mpyr][Ntf_2]$, $[C_{101}mpyr][FSI]$, and $[C_{101}mpyr][Ntf_2]$ were synthesized according to the literature procedure²⁶ (see the ESI†). All the ionic liquids and MX salts were dried *in vacuo* prior to use. The water content of the samples, as determined by Karl Fischer titration, was less than 50 ppm. The solubility was checked at room temperature by mixing a given amount of each MX salt into the ionic liquids in a glove box filled with dried argon gas. The mixtures were stirred for 24 h. If the mixtures give a clear solution, a given amount of each MX salt was added into the solution. This procedure was repeatedly conducted until a turbid solution was obtained.

Results and discussion

Energetic difference between lithium and sodium salts

CCSD(T) interaction energies of the most stable configurations of LiX and NaX salts presented in Fig. 2 clearly indicate an energetic difference between the two alkali ions. It is not surprising that the interaction energy is strongly anion dependent due to the varying polarizability of anions, spanning a wide range of -610 kJ mol^{-1} for $LiFSI$ in the gas phase to -440 kJ mol^{-1} for $NaDCA$ in DMSO. Depending on the method of optimization (be it gas phase or CPCM), LiX salts consistently exhibit much

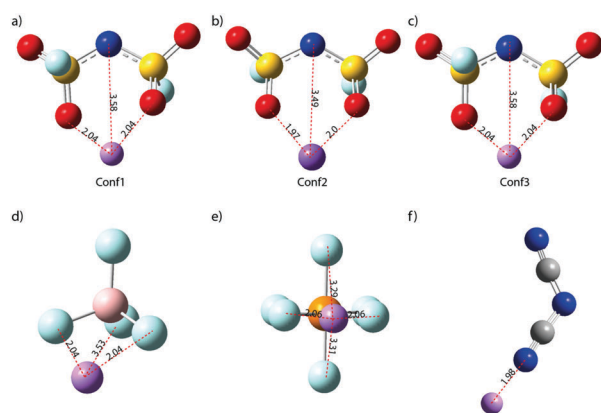


Fig. 1 Optimized structures of LiX complexes in DMSO: (a–c) $LiFSI$, (d) $LiBF_4$, (e) $LiPF_6$ and (f) $LiDCA$.

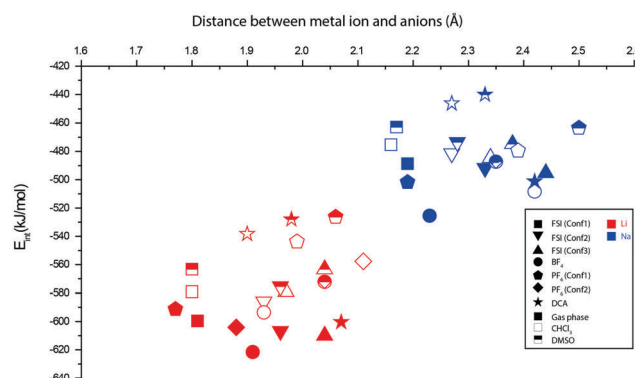


Fig. 2 Inter-ionic distances (R , Å) and CCSD(T) full core interaction energies (E_{INT}) of LiX and NaX (where $X = FSI$, BF_4^- , PF_6^- and DCA).

stronger binding than NaX salts by 88 to 104 kJ mol⁻¹ on average when combined with the same anion. These larger differences cannot be solely attributed to the greater charge density of the lithium cation, which consequently results in stronger electrostatic interactions. The most striking finding from the optimized structures between the metal cation and the ionic liquid anion is the interionic distance, defined as the distance between the metal ion and the closest electronegative atom on the anion. In the case of LiX salts, the distance appears to be very short, falling around 1.86 Å in the gas phase and 1.98 Å in both solvents. Similar distances are normally observed in hydrogen-bonded complexes. In contrast to this trend, the sodium cation adopts a longer separation from the anion, falling between 2.27 and 2.44 Å in either chloroform or DMSO-optimized structures. Considering the effective ionic radii²⁷ of Li⁺ (0.76 Å) and Na⁺ (1.02 Å) and van der Waals radii of atoms in close contact in the optimized structures, the interionic distances in LiX salts are well within the sum of van der Waals radii by at least 0.3 Å, whereas for NaX salts the sum falls within 0.1 Å of the predicted distances. Therefore, it is suggested that additional attractive bonding occurs in complexes of lithium salts apart from the traditional electrostatic interaction. Energetic differences up to 104 kJ mol⁻¹ and shorter than the sum of van der Waals radii interionic distances would imply the presence of covalent-type bonding in lithium salts.

Importance of 1s² electrons of Li⁺ in bonding

It is hypothesized that the 1s² electrons in the lithium cation participate in the bonding with the ionic liquid anion. It is usually not surprising that due to the polarization effects present in a single ion pair of ionic liquids, the orbital energies of the anion consistently decrease within the range of 0.15 eV, whereas orbital energies of the cation increase by the same amount. In the case of LiX and NaX ion pairs, an interesting trend is observed. In the DMSO optimized structures, the orbital energies of the 2s² electrons increase by 0.20 eV in NaX salts, whereas a larger increase of up to 0.27 eV is consistently observed for the 1s² orbital of LiX salts (see Fig. 3). Slightly larger differences are observed for gas phase and chloroform optimized structures, with the largest shifts occurring for LiFSI and NaFSI salts regardless of the solvent polarity. The increased orbital energy to -2.5 eV allows for the 1s² electrons on lithium to actively participate in the bonding between the metal cation and anion.

Since the metal-anion complexes in the gas phase and individual metal cation calculations performed in either DMSO or chloroform produce similar shifts in energy for the 1s² orbital, it is more likely that the effect in LiX and NaX salts originates from pure polarisation created by the anion field, reflecting the polarizability of the anions.^{28,29} To estimate the extent of this polarization effect, the difference between Hartree-Fock (HF) energies in the gas phase and CPCM for the individual lithium cation were calculated to be -406 and -504 kJ mol⁻¹ in CHCl₃ and DMSO, respectively. In contrast to this trend, the analogous comparison for the sodium cation was found to be -334 kJ mol⁻¹ in CHCl₃ and -415 kJ mol⁻¹ in DMSO. Therefore, it can be easily derived that the relatively large shift in

the 1s² orbital causes the HF energy of the lithium cation to decrease by an additional 72 kJ mol⁻¹ in CHCl₃ and 89 kJ mol⁻¹ in DMSO. These findings explain that a large portion of the interaction energy difference between LiX and NaX salts comes from the involvement of the 1s² electrons on lithium in the bonding with the anion. Based on the orbital energy of the 1s² electrons in lithium, the following order of increasing covalency can be concluded: FSI > BF₄⁻ ≈ DCA > PF₆⁻.

Positive correlation interaction energy and basis set effect

In quantum-chemical calculations the 1s² electrons on the lithium cation and 2s² electrons of the sodium cation would be considered core electrons due to their low (*i.e.* more negative) orbital energy. For NaX salts the orbital energy of -1.6 eV for the 2p⁶ electrons on Na⁺ would automatically guarantee their inclusion in the valence space by default in any correlated calculation (see Fig. 3). Together with the frozen-core CCSD(T) calculations, full-core calculations were also performed, in which the 1s² electrons on Li⁺ and the 2s² electrons on Na⁺ were included in the valence space. Electron correlation was separated from the HF energy and correlation interaction energies were calculated as shown in Fig. 4 for the FSI-based salts. Positive correlation interaction energies are observed for all configurations of LiFSI in the CCSD(T) frozen core calculations falling between 1.2 and 9.6 kJ mol⁻¹, far beyond the accuracy of these calculations. Once

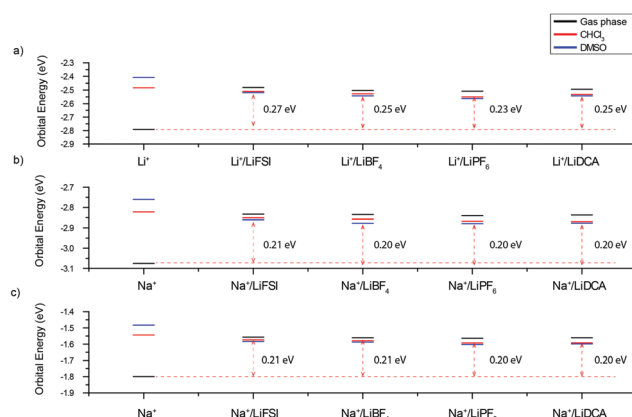


Fig. 3 Orbital energies (in eV) of MX: (a) 1s² orbital energy of LiX, (b) 2s² orbital energy of NaX and (c) 2p⁶ orbital energies of NaX (where X = FSI, BF₄⁻, PF₆⁻ and DCA).

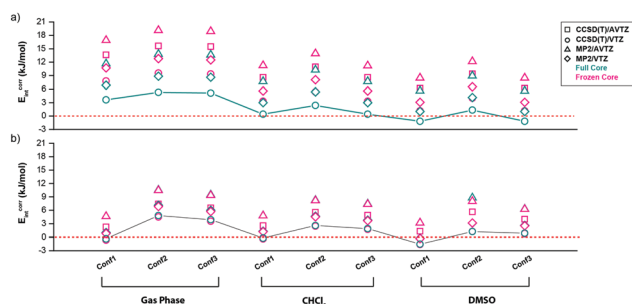


Fig. 4 Correlation interaction energy (*E* in kJ mol⁻¹) of (a) LiFSI and (b) NaFSI. The red dashed line of 0 kJ mol⁻¹ is only advisory.

the $1s^2$ electrons on Li^+ were included as valence, the correlation interaction energy decreased, approaching negative values for the DMSO-optimized structures. There is a consistent difference between the full-core and frozen-core calculations by 4.2 kJ mol^{-1} for gas phase structures, 2.9 kJ mol^{-1} for chloroform and 2.5 kJ mol^{-1} for DMSO, with decreasing difference reflecting the effect of increasing interionic distance with solvent polarity. These results clearly demonstrate the engagement of the $1s^2$ electrons in the bonding between the lithium cation and the anion, affecting the correlation energy and thus making the bonding partially covalent. Due to the pull of the $1s^2$ electrons towards the anion, the interionic distance decreases followed by increasing electrostatic interaction and hence, overall interaction energy regardless of the anion type.

In the case of NaFSI, no applicable differences (usually within 0.1 kJ mol^{-1}) were observed between full-core and frozen-core calculations, highlighting the fact that the $2s^2$ orbital did not participate in the bonding. The correlation interaction energy falls in a very narrow range of -1.5 to 2.6 kJ mol^{-1} for CPCM-optimized structures, which is well within the accuracy of these calculations. The rest of the studied ionic liquid anions further confirmed the above observation (see Fig. 5). In LiX salts, the inclusion of the $1s^2$ electrons was necessary for the interaction correlation energy to become negative and only slightly positive within the systematic error of the CCSD(T)/aVTZ combination, whereas for NaX salts the correlation interaction energy did not change with the inclusion of the $2s^2$ electrons. Slightly positive values of correlation interaction energies within the range of 1 kJ mol^{-1} are observed for DCA-based salts in CPCM-optimized structures. It has to be pointed out that the involvement of $1s^2$ electrons in bonding of LiX salts clearly highlights the need for a highly correlated level of theory such as CCSD(T) with a much larger basis set of $5-\zeta$ quality to properly recover electron correlation effects in LiX salts and to some extent in NaX salts. The full-core MP2 method usually produces a positive correlation interaction energy up to 19 kJ mol^{-1} when combined with either cc-pVTZ or aug-cc-pVTZ depending on the ion pair configuration and optimization method (see Fig. 4 and 5).

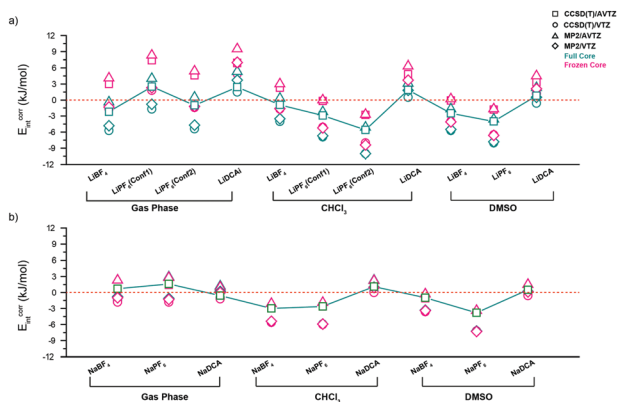


Fig. 5 Correlation interaction energy ($E_{\text{INT}}^{\text{corr}}$ in kJ mol^{-1}) of (a) LiX salts and (b) NaX salts (where $X = \text{BF}_4^-$, PF_6^- and DCA). The red dashed line of 0 kJ mol^{-1} is only advisory.

Extrapolation of MP2 correlation energy to the complete basis set limit seems to help capture the electron correlation producing values within 2 kJ mol^{-1} of the benchmark values (see the ESI†). The presented data indicate that MP2/CBS might be capable of recovering to electron correlation for the $1s^2$ electrons on Li^+ as well as $2p^6$ electrons on Na^+ . To this end, the mystery surrounding the positive electron correlation energy in LiX salts has now been explained.

To further confirm the importance of inclusion of larger basis sets at highly correlated levels of theory, a series of Dunning's basis sets, aug-cc-pVTZ, cc-pVQZ, aug-cc-pVQZ and cc-pv5Z were used to perform full core CCSD(T) calculations on DMSO-optimized LiX and NaX complexes with $X = \text{F}$ and Cl . For the fluorides, the correlation interaction energy consistently decreased with increasing basis set from 0.5 to $-10.5 \text{ kJ mol}^{-1}$ for LiF and from 1.5 to $-11.2 \text{ kJ mol}^{-1}$ for NaF. Similar trends are observed for LiCl and NaCl, with the correlation interaction energy values decreasing from -1.4 to -5.7 kJ mol^{-1} for LiCl and from -1.2 to -7.0 kJ mol^{-1} for NaCl. These data clearly identify the need for larger basis sets to fully recover electron correlation in these systems.

Charge transfer between cation and anion

There is a third effect occurring in these systems manifesting itself through the charge transfer of electron density from the anion to the metal cation that also results in increased covalency. This phenomenon has been discussed and confirmed in ionic liquids, resulting in the reduction of the charges on ions and subsequently a decrease in their cohesive (*i.e.* electrostatic) energy.^{28,30,31} The latter allows for faster ionic dynamics and is in good agreement with the experimental data including conductivity and viscosity of ionic liquids. Previously we showed that the Geodesic scheme was robust in capturing the extent of charge transfer in ionic liquid ions.²⁸ It has to be noted that the Natural Bond Population analysis would not produce charge transfer in ion pairs of inorganic salts and ionic liquids for fundamental reasons. In LiX and NaX salts, similar values for the net charge transfer were observed falling between 0.05 and 0.17 , with the LiX salts typically having a larger net charge transfer due to shorter interionic distances (see Fig. 6). Among the NaX salts, the lowest energy conformation of NaFSI (labeled as conf3 in Fig. 6) had the largest net charge transfer. The charge transfer effect is not as drastic as the involvement of the $1s^2$ orbitals in the bonding, since LiX salts are still more strongly bound than NaX salts by at least 90 kJ mol^{-1} .

Solubility of lithium and sodium salts in ionic liquids

Why does the increased covalency in LiX salts translate into their better solubility? It is hypothesized that in the crystal structure these salts might be viewed as the packing of "neutral", individual "complexes" with a strong orbital overlap between the lithium cation and the ionic liquid anion (see Fig. 7(a)). From the thermodynamic point of view it is much easier to solvate neutral molecules as these already form stable thermodynamic entities. In contrast to this effect, the solvation of ions from inorganic sodium salts (see Fig. 7(b)) involves not only breaking an ion away

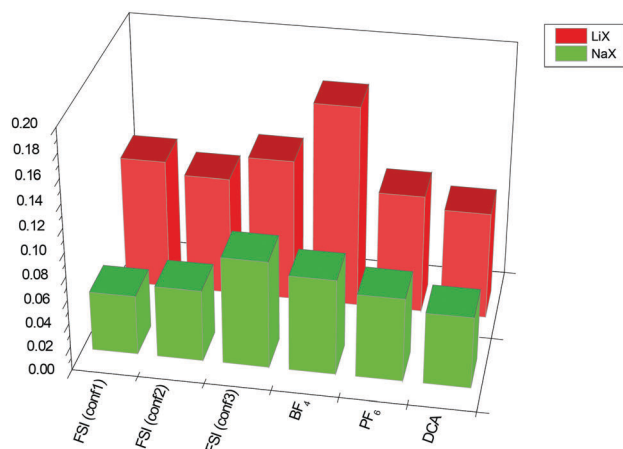


Fig. 6 Net charge transfer between the alkali metal cation and ionic liquid anion in DMSO optimized structures of LiX and NaX salts (where X = BF_4^- , PF_6^- and DCA).

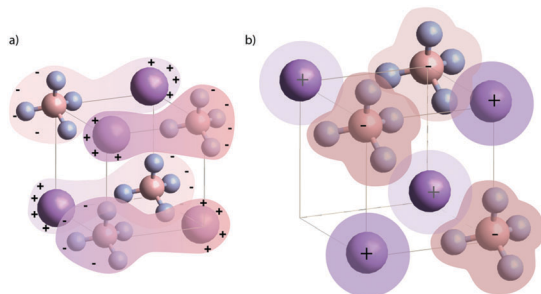


Fig. 7 Proposed bonding in (a) LiX salts and (b) NaX salts.

from a strongly bound lattice but also providing thermodynamic stability to the positive or negative charge on the ion. This is achieved through the creation of solvation shells around each individual ion. Typically, polar solvents with high re-organizational entropy such as water are more likely to easily solvate charged species. It is for this reason that many alkali metal salts have poor solubility in polar organic solvents such as alcohols.³² Although quite polar, ionic liquids do not carry the same concentration of ions with coordinating ability per unit volume as do molecular solvents such as acetonitrile and water and therefore, their organization entropy would be much lower, thus reducing the likelihood of metal ion solvation.

In this study, solubility measurements of MX salts (where M = Li^+ and Na^+ and X = NTf_2^- , BF_4^- and PF_6^-) were performed in 6 different ionic liquids (see Fig. 8). It is clearly seen that LiNTf_2 (> 2 M) dissolved much better than the sodium analogue (< 2 M). The ionic liquids with the FSI anion exhibited a higher solubility as compared with that of NTf_2^- -based ionic liquids for both LiNTf_2 and NaNTf_2 , with the concentration of LiNTf_2 reaching about 6 M in $[\text{C}_2\text{mim}][\text{FSI}]$. In the case of the BF_4^- and PF_6^- anions, LiX salts have at least an order of magnitude better solubility than NaX salts, with the latter exhibiting poor solubility < 0.1 M. It does not appear that there is a strong ionic liquid cation or anion effect on the solubility of LiX salts, with

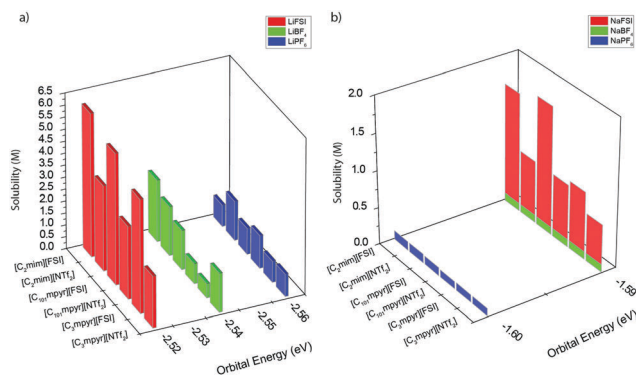


Fig. 8 The solubility of MX in different ionic liquids: (a) LiX and (b) NaX (where X = FSI, BF_4^- and PF_6^-).

the FSI-based ionic liquids only slightly outperforming the NTf_2^- ones. In the case of NaX salts, NaNTf_2 showed good solubility in all ionic liquids studied similar to that of LiPF_6 . The $1s^2$ orbital energy on Li^+ and $2s^2$ orbital energy on Na^+ of the most energetically stable M^+X^- configurations in DMSO are plotted against solubility numbers in Fig. 7. In their chemical structure, the NTf_2^- and FSI anions are very similar and therefore, the orbital energies of LiFSI and NaFSI were used to represent LiNTf_2 and NaNTf_2 , respectively. One can clearly see that with increasing orbital energy (and hence decreasing covalency) the solubility of LiX salts consistently decreases following the previously established order: $\text{NTf}_2^- > \text{BF}_4^- > \text{PF}_6^-$. Among the studied NaX salts, NaFSI was the only one that showed appreciable solubility in ionic liquids due to the presence of covalency due to increased charge transfer. More ionic NaBF_4 and NaPF_6 followed the expected trend of poor solubility. It is also suggested that entropic effects might play an important role in the increased solubility of LiFSI and NaFSI compared to salts coupled with BF_4^- and PF_6^- anions. For example, the dissolution of lithium and sodium trifluoromethanesulfonates (OTf) in $[\text{C}_4\text{mim}][\text{OTf}]$ was facilitated by a slightly positive change in entropy of dissolution.³³ On the other hand, sodium halides were all found to exhibit slightly negative change in entropy upon dissolution in the same ionic liquid.

Conclusions

In summary, LiX salts have been shown to dissolve much better in ionic liquids than their sodium analogues. The solvated lithium salt is more likely to continue to exist as neutral close-contact complexes with a further possibility of forming stable LiX_n^{n-1} complexes.³⁴ This finding is in good agreement with observations of significantly decreased conductivity and increased viscosity of ionic liquids upon addition of lithium salts. From the theoretical point of view, the lithium cation cannot be treated as a point charge in molecular dynamics simulations, with the $1s^2$ electrons contributing close to 90 kJ mol^{-1} to the stability of the bonding in LiX salts compared. For the complete capture of electron correlation effects in these salts, larger basis sets of at least a $5-\zeta$ quality are

needed when combined with CCSD(T). Appreciable net charge transfer between the lithium cation and anion also needs to be taken into account for reliable predictions of their properties.

To overcome issues of solubility of sodium salts, ionic liquids with strong coordination ability are recommended. Some of the strategies include oligo-ether units in the side chain of the cation, strong Lewis basic anions and mixtures of ILs and solvents (e.g. glyme³⁵) or additives (e.g. zwitterions³⁶) with strong coordinating ability and high organization entropy. Once the solubility issues are overcome, sodium cations are more likely to be integrated into the ionic environment resulting in better transport properties through the ionic medium and hence resulting in better efficiencies of sodium-ion batteries. Among ionic liquid anions, BF_4^- , PF_6^- and DCA are highly recommended due to decreased covalency.

Acknowledgements

The authors gratefully acknowledge a generous allocation of computer resources through the Monash eResearch Centre and the National Computational Infrastructure. This work is generously supported by the Australian Research Council through a Discovery Project and a Future Fellowship for E. I. I.

Notes and references

- 1 A. Eftekhari, Y. Liu and P. Chen, *J. Power Sources*, 2016, **334**, 221.
- 2 I. Hasa, S. Passerini and J. Hassoun, *J. Power Sources*, 2016, **303**, 203.
- 3 M. Moreno, E. Simonetti, G. B. Appetecchi, M. Carewska, M. Montanino, G.-T. Kim, N. Loeffler and S. Passerini, *J. Electrochem. Soc.*, 2017, **164**, A6026.
- 4 H. Srour, L. Chancelier, E. Bolimowska, T. Gutel, S. Mailley, H. Rouault and C. C. Santini, *J. Appl. Electrochem.*, 2016, **46**, 149.
- 5 K. Vignarooban, R. Kushagra, A. Elango, P. Badami, B. E. Mellander, X. Xu, T. G. Tucker, C. Nam and A. M. Kannan, *Int. J. Hydrogen Energy*, 2016, **41**, 2829.
- 6 S. N. Chavan, A. Tiwari, T. C. Nagaiah and D. Mandal, *Phys. Chem. Chem. Phys.*, 2016, **18**, 16116.
- 7 G. A. Elia, U. Ulissi, S. Jeong, S. Passerini and J. Hassoun, *Energy Environ. Sci.*, 2016, **9**, 3210.
- 8 M. Armand, F. Endres, D. R. Macfarlane, H. Ohno and B. Scrosati, *Nat. Mater.*, 2009, **8**, 621.
- 9 D. R. Macfarlane, M. Forsyth, P. C. Howlett, M. Kar, S. Passerini, J. M. Pringle, H. Ohno, M. Watanabe, F. Yan, W. Zheng, S. Zhang and J. Zhang, *Nat. Rev. Mater.*, 2016, **1**, 15005.
- 10 P. E. Ross, *Boeing's Battery Blues*, North American, 2013.
- 11 H. Matsumoto, H. Sakaebe, K. Tatsumi, M. Kikuta, E. Ishiko and M. Kono, *J. Power Sources*, 2006, **160**, 1308.
- 12 S. Seki, Y. Ohno, Y. Kobayashi, H. Miyashiro, A. Usami, Y. Mita, H. Tokuda, M. Watanabe, K. Hayamizu, S. Tsuzuki, M. Hattori and N. Terada, *J. Electrochem. Soc.*, 2007, **154**, A173.
- 13 K. Hayamizu, Y. Aihara, H. Nakagawa, T. Nukuda and W. S. Price, *J. Phys. Chem. B*, 2004, **108**, 19527.
- 14 A. Guerfi, M. Dontigny, P. Charest, M. Petitclerc, M. Lagacé, A. Vijh and K. Zaghib, *J. Power Sources*, 2010, **195**, 845.
- 15 M. Shimizu, H. Usui and H. Sakaguchi, *Phys. Chem. Chem. Phys.*, 2016, **18**, 5139.
- 16 A. B. Pereiro, J. M. M. Araújo, F. S. Oliveira, J. M. S. S. Esperança, J. N. Canongia Lopes, I. M. Marrucho and L. P. N. Rebelo, *J. Chem. Thermodyn.*, 2012, **55**, 29.
- 17 H. Yoon, H. Zhu, A. Hervault, M. Armand, D. R. Macfarlane and M. Forsyth, *Phys. Chem. Chem. Phys.*, 2014, **16**, 12350.
- 18 P. Su and H. Li, *J. Chem. Phys.*, 2009, **131**, 014102.
- 19 M. Cossi, G. Scalmani, N. Rega and V. Barone, *J. Chem. Phys.*, 2002, **117**, 43.
- 20 S. F. Boys and F. Bernardi, *Mol. Phys.*, 1970, **19**, 553.
- 21 T. Helgaker, P. Jorgensen and J. Olsen, *Molecular Electronic-Structure Theory*, Wiley, 2000, p. 908.
- 22 M. A. Spackman, *J. Comput. Chem.*, 1996, **17**, 1.
- 23 M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H. P. Hratchian, A. F. Izmaylov, J. Bloino, G. Zheng, J. L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J. A. Montgomery Jr., J. E. Peralta, F. Ogliaro, M. J. Bearpark, J. Heyd, E. N. Brothers, K. N. Kudin, V. N. Staroverov, R. Kobayashi, J. Normand, K. Raghavachari, A. P. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, N. Rega, N. J. Millam, M. Klene, J. E. Knox, J. B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R. E. Stratmann, O. Yazyev, A. J. Austin, R. Cammi, C. Pomelli, J. W. Ochterski, R. L. Martin, K. Morokuma, V. G. Zakrzewski, G. A. Voth, P. Salvador, J. J. Dannenberg, S. Dapprich, A. D. Daniels, Ö. Farkas, J. B. Foresman, J. V. Ortiz, J. Cioslowski and D. J. Fox, *Gaussian 09*, Gaussian Inc., Wallingford, CT, USA, 2009.
- 24 J. M. Turney, A. C. Simmonett, R. M. Parrish, E. G. Hohenstein, F. A. Evangelista, J. T. Fermann, B. J. Mintz, L. A. Burns, J. J. Wilke, M. L. Abrams, N. J. Russ, M. L. Leininger, C. L. Janssen, E. T. Seidl, W. D. Allen, H. F. Schaefer, R. A. King, E. F. Valeev, C. D. Sherrill and T. D. Crawford, *Wiley Interdiscip. Rev.: Comput. Mol. Sci.*, 2012, **2**, 556.
- 25 M. W. Schmidt, K. K. Baldridge, J. A. Boatz, S. T. Elbert, M. S. Gordon, J. H. Jensen, S. Koseki, N. Matsunaga, K. A. Nguyen, S. Su, T. L. Windus, M. Dupuis and J. A. Montgomery, *J. Comput. Chem.*, 1993, **14**, 1347.
- 26 S. Yamaguchi, M. Yoshizawa-Fujita, Y. Takeoka and M. Rikukawa, *J. Power Sources*, 2016, **331**, 308.
- 27 A. Bondi, *J. Phys. Chem.*, 1964, **68**, 441.
- 28 J. Rigby and E. I. Izgorodina, *Phys. Chem. Chem. Phys.*, 2013, **15**, 1632.
- 29 E. I. Izgorodina, M. Forsyth and D. R. Macfarlane, *Phys. Chem. Chem. Phys.*, 2009, **11**, 2452.

- 30 E. I. Izgorodina, *Phys. Chem. Chem. Phys.*, 2011, **13**, 4189.
- 31 E. I. Izgorodina, Z. L. Seeger, D. L. A. Scarborough and S. Y. S. Tan, *Chem. Rev.*, 2017, **117**, 6696.
- 32 E. Hawlicka, *Z. Naturforsch.*, 1986, **41**, 939.
- 33 O. Kuzmina, E. Bordes, J. Schmauck, P. A. Hunt, J. P. Hallett and T. Welton, *Phys. Chem. Chem. Phys.*, 2016, **18**, 16161.
- 34 C. W. Bauschlicher, J. B. Haskins, E. W. Bucholz, J. W. Lawson and O. Borodin, *J. Phys. Chem. B*, 2014, **118**, 10785.
- 35 K. Yoshida, M. Nakamura, Y. Kazue, N. Tachikawa, S. Tsuzuki, S. Seki, K. Dokko and M. Watanabe, *J. Am. Chem. Soc.*, 2011, **133**, 13121.
- 36 S. Yamaguchi, M. Yoshizawa-Fujita, Y. Takeoka and M. Rikukawa, *J. Power Sources*, 2016, **331**, 308.